

Class8 - Breast Cancer Mini Project

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The goal of this mini-project is for you to explore a complete analysis using the unsupervised learning techniques covered in class. You'll extend what you've learned by combining PCA as a preprocessing step to clustering using data that consist of measurements of cell nuclei of human breast masses. This expands on our RNA-Seq analysis from last day. The data itself comes from the Wisconsin Breast Cancer Diagnostic Data Set first reported by K. P. Benne and O. L. Mangasarian: "Robust Linear Programming Discrimination of Two Linearly Inseparable Sets". Values in this data set describe characteristics of the cell nuclei present in digitized images of a fine needle aspiration (FNA) of a breast mass.

Data Import

```
wisc.df <- read.csv("WisconsinCancer.csv", row.names=1)
head(wisc.df)
```

	diagnosis	radius_mean	texture_mean	perimeter_mean	area_mean
842302	M	17.99	10.38	122.80	1001.0
842517	M	20.57	17.77	132.90	1326.0
84300903	M	19.69	21.25	130.00	1203.0
84348301	M	11.42	20.38	77.58	386.1

84358402	M	20.29	14.34	135.10	1297.0
843786	M	12.45	15.70	82.57	477.1
	smoothness_mean	compactness_mean	concavity_mean	concave.points_mean	
842302	0.11840	0.27760	0.3001	0.14710	
842517	0.08474	0.07864	0.0869	0.07017	
84300903	0.10960	0.15990	0.1974	0.12790	
84348301	0.14250	0.28390	0.2414	0.10520	
84358402	0.10030	0.13280	0.1980	0.10430	
843786	0.12780	0.17000	0.1578	0.08089	
	symmetry_mean	fractal_dimension_mean	radius_se	texture_se	perimeter_se
842302	0.2419	0.07871	1.0950	0.9053	8.589
842517	0.1812	0.05667	0.5435	0.7339	3.398
84300903	0.2069	0.05999	0.7456	0.7869	4.585
84348301	0.2597	0.09744	0.4956	1.1560	3.445
84358402	0.1809	0.05883	0.7572	0.7813	5.438
843786	0.2087	0.07613	0.3345	0.8902	2.217
	area_se	smoothness_se	compactness_se	concavity_se	concave.points_se
842302	153.40	0.006399	0.04904	0.05373	0.01587
842517	74.08	0.005225	0.01308	0.01860	0.01340
84300903	94.03	0.006150	0.04006	0.03832	0.02058
84348301	27.23	0.009110	0.07458	0.05661	0.01867
84358402	94.44	0.011490	0.02461	0.05688	0.01885
843786	27.19	0.007510	0.03345	0.03672	0.01137
	symmetry_se	fractal_dimension_se	radius_worst	texture_worst	
842302	0.03003	0.006193	25.38	17.33	
842517	0.01389	0.003532	24.99	23.41	
84300903	0.02250	0.004571	23.57	25.53	
84348301	0.05963	0.009208	14.91	26.50	
84358402	0.01756	0.005115	22.54	16.67	
843786	0.02165	0.005082	15.47	23.75	
	perimeter_worst	area_worst	smoothness_worst	compactness_worst	
842302	184.60	2019.0	0.1622	0.6656	
842517	158.80	1956.0	0.1238	0.1866	
84300903	152.50	1709.0	0.1444	0.4245	
84348301	98.87	567.7	0.2098	0.8663	
84358402	152.20	1575.0	0.1374	0.2050	
843786	103.40	741.6	0.1791	0.5249	
	concavity_worst	concave.points_worst	symmetry_worst		
842302	0.7119	0.2654	0.4601		
842517	0.2416	0.1860	0.2750		
84300903	0.4504	0.2430	0.3613		
84348301	0.6869	0.2575	0.6638		
84358402	0.4000	0.1625	0.2364		

843786	0.5355	0.1741	0.3985
	fractal_dimension_worst		
842302	0.11890		
842517	0.08902		
84300903	0.08758		
84348301	0.17300		
84358402	0.07678		
843786	0.12440		

the first column here `wisc.df$diagnosis` is a pathologist provided expert diagnosis. We will not be using this for our unsupervised analysis as it is essentially the “answer” to the question which cell samples are malignant or benign.

To make sure we don’t accidentally include this in our analysis, lets create a new data.frame that omits this first column

```
wisc.df$diagnosis
```

```
[1] "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M"
[19] "M" "B" "B" "B" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M"
[37] "M" "B" "M" "M" "M" "M" "M" "M" "M" "M" "B" "M" "B" "B" "B" "B" "M"
[55] "M" "B" "M" "M" "B" "B" "B" "B" "M" "B" "M" "M" "B" "B" "B" "B" "M"
[73] "M" "M" "B" "M" "B" "M" "M" "B" "B" "B" "M" "M" "B" "M" "M" "M" "B"
[91] "B" "M" "B" "B" "M" "M" "B" "B" "B" "M" "M" "B" "B" "B" "B" "M" "B"
[109] "M" "B" "B" "B" "B" "B" "B" "B" "B" "M" "M" "M" "B" "M" "M" "B" "B"
[127] "M" "M" "B" "M" "B" "M" "M" "B" "M" "M" "B" "B" "M" "B" "B" "M" "B"
[145] "B" "B" "M" "B" "B" "B" "B" "B" "B" "B" "B" "B" "M" "B" "B" "B" "M"
[163] "M" "B" "M" "B" "B" "M" "M" "B" "B" "M" "M" "B" "B" "B" "B" "M" "B"
[181] "M" "M" "M" "B" "M" "B" "M" "B" "B" "B" "M" "B" "B" "M" "M" "B" "M"
[199] "M" "M" "B" "M" "M" "M" "B" "M" "B" "M" "B" "B" "M" "B" "M" "M" "M"
[217] "B" "B" "M" "M" "B" "B" "B" "M" "B" "B" "B" "B" "B" "M" "M" "B" "B"
[235] "B" "B" "M" "M" "B" "M" "B" "B" "B" "B" "M" "B" "B" "B" "B" "B" "M"
[253] "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "M" "B" "B" "B"
[271] "B" "B" "M" "B" "M" "B" "B" "M" "B" "B" "M" "B" "M" "M" "B" "B" "B"
[289] "B" "B" "B" "B" "B" "B" "B" "B" "B" "M" "B" "B" "M" "B" "M" "B" "B"
[307] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "M" "B" "B" "B" "M" "B"
[325] "B" "B" "B" "B" "M" "M" "M" "B" "B" "B" "B" "M" "B" "M" "B" "M" "B"
[343] "B" "M" "B" "B" "B" "B" "B" "B" "B" "M" "M" "M" "B" "B" "B" "B" "B"
[361] "B" "B" "B" "B" "B" "M" "M" "B" "M" "M" "M" "B" "M" "M" "B" "B" "B"
[379] "B" "M" "B" "B" "B" "B" "B" "M" "B" "B" "B" "M" "B" "B" "M" "M" "B"
[397] "B" "B" "B" "B" "M" "B" "B" "B" "B" "B" "B" "B" "M" "B" "B" "B" "B"
[415] "M" "B" "B" "M" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "M"
```

```
[433] "M" "M" "B" "M" "B" "B" "B" "B" "B" "M" "B" "B" "M" "B" "M" "B" "B" "M"
[451] "B" "M" "B" "B" "B" "B" "B" "B" "B" "B" "M" "M" "B" "B" "B" "B" "B" "B"
[469] "M" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "M" "B" "B" "B" "B" "B" "B"
[487] "B" "M" "B" "M" "B" "B" "M" "B" "B" "B" "B" "B" "M" "M" "B" "M" "B" "M"
[505] "B" "B" "B" "B" "B" "M" "B" "B" "M" "B" "M" "B" "M" "M" "B" "B" "B" "M"
[523] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "M" "B" "M" "M" "B" "B" "B"
[541] "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B" "B"
[559] "B" "B" "B" "B" "M" "M" "M" "M" "M" "M" "M" "B"
```

```
wisc.data<-wisc.df[,-1]
```

Store the diagnosis as a vector for us later when we compare our results to those from experts in the field.

```
# Create diagnosis vector for later
diagnosis <-factor(wisc.df$diagnosis)
```

Q1. How many observations are in this dataset?

There are 569 observation patients in the dataset.

Q2. How many of the observations have a malignant diagnosis?

```
table(wisc.df$diagnosis)
```

```
  B   M
357 212
```

Q3. How many variables/features in the data are suffixed with `_mean`?

```
#colnames(wisc.data)
length(grep("_mean",colnames(wisc.data)))
```

```
[1] 10
```

##Principle Component Analysis (PCA)

The `prcomp()` function to do PCA has a `scale=FALSE` default. In general we nearly always want to set this to `TRUE` so our analysis is not dominated by columns/variables in our data set that have high standard deviation and mean when compared to others just because the units of measurement are on different scales.

```
wisc.pr <- prcomp(wisc.data, scale=TRUE)
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

Q4. From your results, what proportion of the original variance is captured by the first principal components (PC1)?

PC1 Proportion of Variance = 0.4427 PC1 alone explains about 44.27% of the total variance in the dataset.

Q5. How many principal components (PCs) are required to describe at least 70% of the original variance in the data?

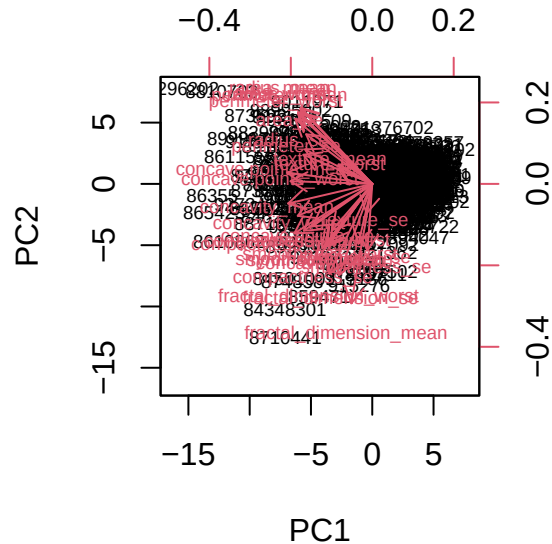
PC1: 0.4427 (44%) PC1+PC2: 0.6324(63%) PC1+PC2+PC3: 0.72636(72%) the first 3 is going to exceed the 70% of the original variance.

Q6. How many principal components (PCs) are required to describe at least 90% of the original variance in the data?

until PC6: 0.88759 (88%) until PC7: 0.91010 (91%) the first 7 principal components to exceed 90% of the variance.

Interpreting PCA results

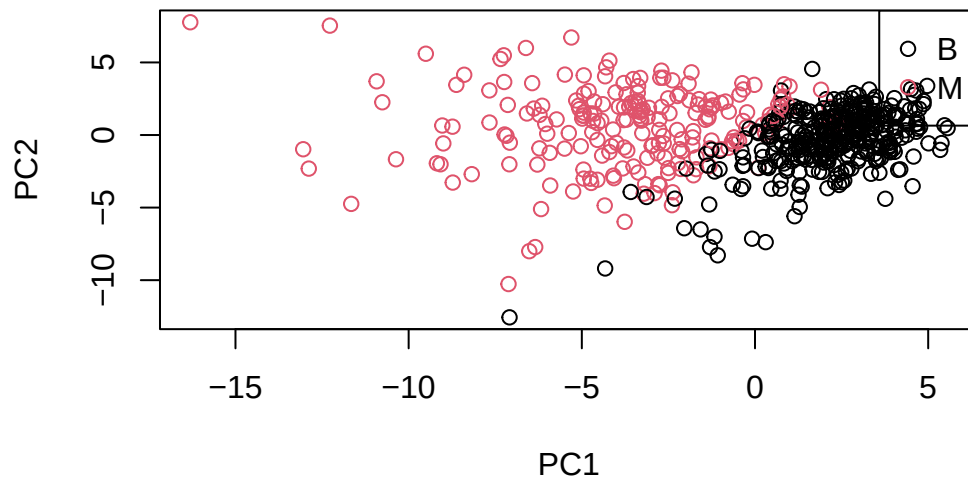
```
biplot(wisc.pr, scale = 0, cex = 0.6)
```



Q7. What stands out to you about this plot? Is it easy or difficult to understand? Why?

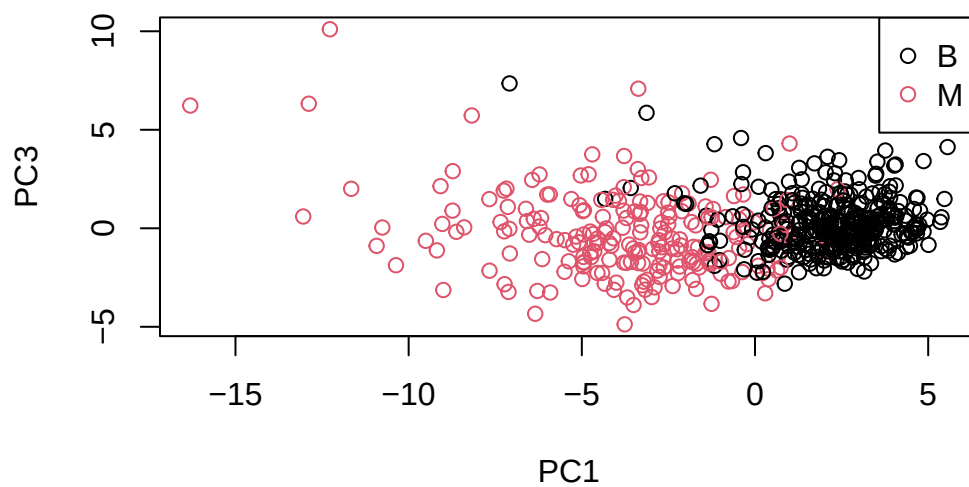
because there are so many variables and data points, the plot is cluttered and difficult to interpret clearly. Biplots are best suited for datasets with a small number of variables, so for this dataset, it's challenging to visually extract meaningful structure.

```
# Scatter plot: PC1 vs PC2 colored by diagnosis
plot(wisc.pr$x[, 1:2], col = diagnosis,
     xlab = "PC1", ylab = "PC2")
legend("topright", legend = levels(diagnosis),
     col = 1:length(levels(diagnosis)), pch = 1)
```



Q8. Generate a similar plot for principal components 1 and 3. What do you notice about these plots?

```
# Scatter plot: PC1 vs PC3 colored by diagnosis
plot(wisc.pr$x[, c(1, 3)], col = diagnosis,
     xlab = "PC1", ylab = "PC3")
legend("topright", legend = levels(diagnosis),
     col = 1:length(levels(diagnosis)), pch = 1)
```

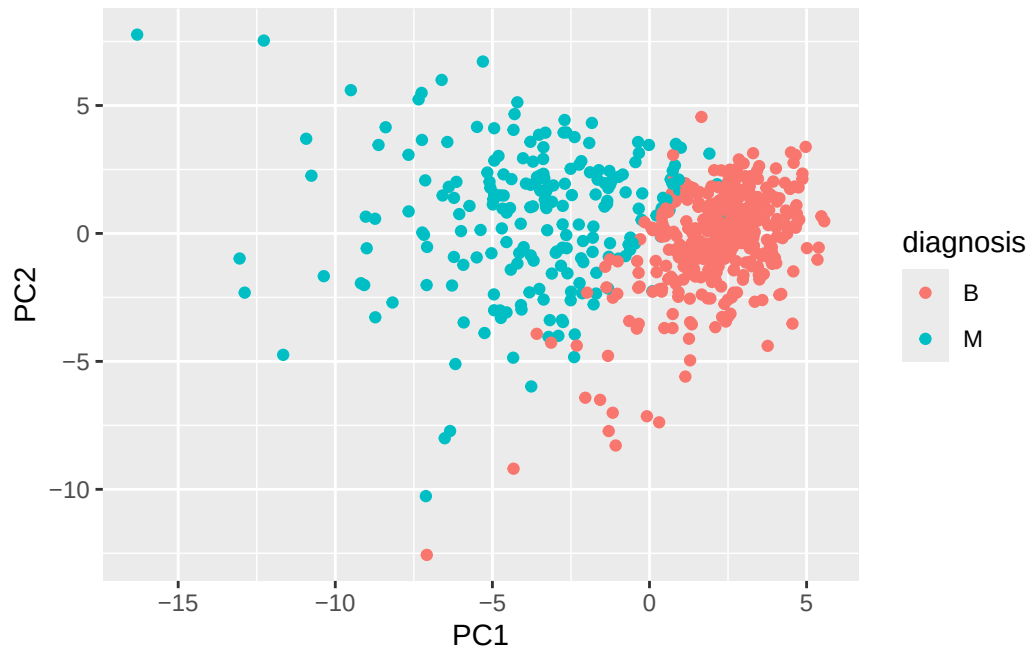


PC1 carries most of the class-separating signal; PC2 contributes a bit, while PC3 contributes comparatively less to separation.

The main PCA plot

The main PC result figure is called a “score plot” or “PC plot” or “ordination plot”...

```
library(ggplot2)
ggplot(wisc.pr$x)+
  aes(PC1,PC2, col=diagnosis)+
  geom_point()
```

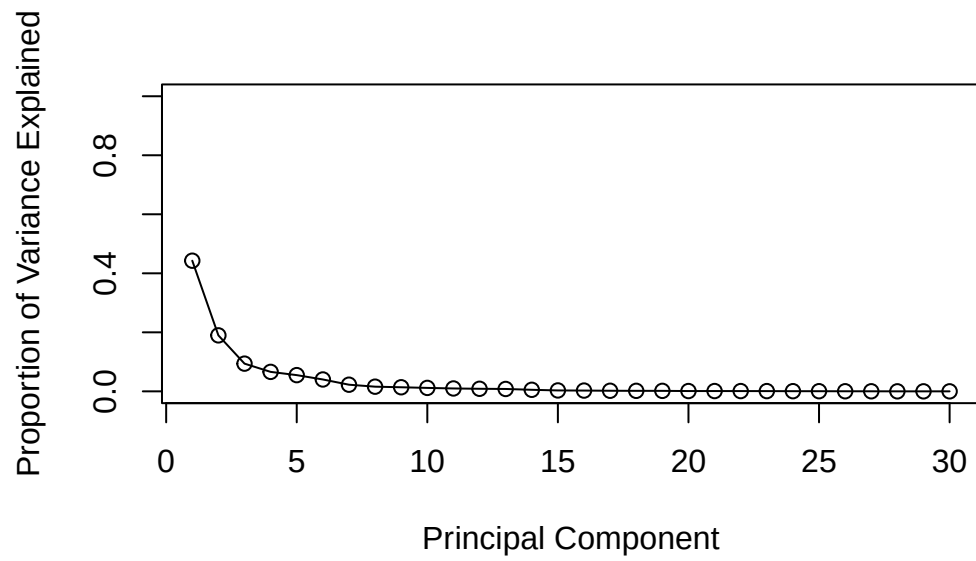
Variance explained

```
# Calculate variance of each component
pr.var <- wisc.pr$sdev^2
head(pr.var)
```

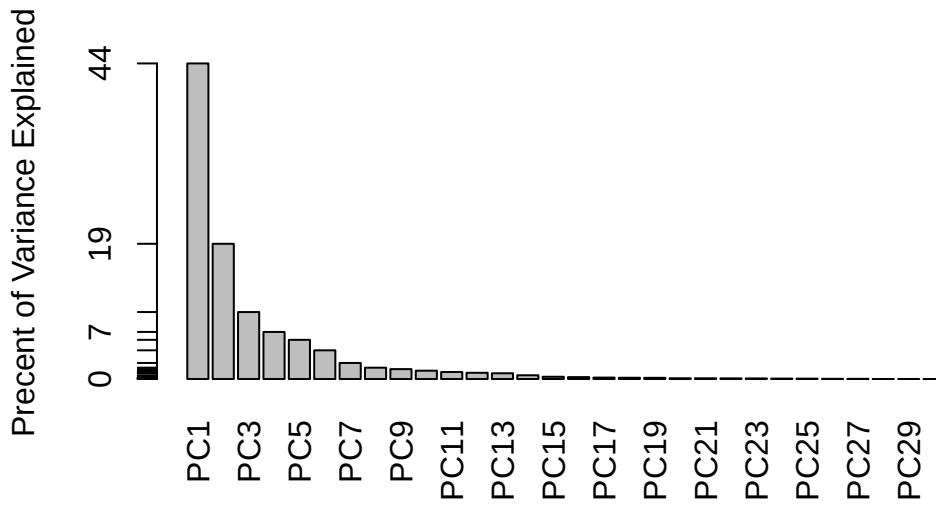
```
[1] 13.281608  5.691355  2.817949  1.980640  1.648731  1.207357
```

```
# Variance explained by each principal component: pve
pve <- pr.var / sum(pr.var)

# Plot variance explained for each principal component
plot(pve, xlab = "Principal Component",
     ylab = "Proportion of Variance Explained",
     ylim = c(0, 1), type = "o")
```



```
# Alternative scree plot of the same data, note data driven y-axis
barplot(pve, ylab = "Precent of Variance Explained",
        names.arg=paste0("PC",1:length(pve)), las=2, axes = FALSE)
axis(2, at=pve, labels=round(pve,2)*100 )
```



###Communicating PCA results

Q9. For the first principal component, what is the component of the loading vector (i.e. `wisc.pr$rotation[,1]`) for the feature `concave.points_mean`?

```
wisc.pr$rotation["concave.points_mean","PC1"]
```

```
[1] -0.2608538
```

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731

Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

The negative sign reflects the orientation of PC1 and does not affect its importance — the magnitude (0.26) shows it's one of the most influential features shaping PC1.

Q10. What is the minimum number of principal components required to explain 80% of the variance of the data?

We need PC5 with the one that is above 80%

```
summary(wisc.pr)
```

Importance of components:

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
Standard deviation	3.6444	2.3857	1.67867	1.40735	1.28403	1.09880	0.82172
Proportion of Variance	0.4427	0.1897	0.09393	0.06602	0.05496	0.04025	0.02251
Cumulative Proportion	0.4427	0.6324	0.72636	0.79239	0.84734	0.88759	0.91010
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
Standard deviation	0.69037	0.6457	0.59219	0.5421	0.51104	0.49128	0.39624
Proportion of Variance	0.01589	0.0139	0.01169	0.0098	0.00871	0.00805	0.00523
Cumulative Proportion	0.92598	0.9399	0.95157	0.9614	0.97007	0.97812	0.98335
	PC15	PC16	PC17	PC18	PC19	PC20	PC21
Standard deviation	0.30681	0.28260	0.24372	0.22939	0.22244	0.17652	0.1731
Proportion of Variance	0.00314	0.00266	0.00198	0.00175	0.00165	0.00104	0.0010
Cumulative Proportion	0.98649	0.98915	0.99113	0.99288	0.99453	0.99557	0.9966
	PC22	PC23	PC24	PC25	PC26	PC27	PC28
Standard deviation	0.16565	0.15602	0.1344	0.12442	0.09043	0.08307	0.03987
Proportion of Variance	0.00091	0.00081	0.0006	0.00052	0.00027	0.00023	0.00005
Cumulative Proportion	0.99749	0.99830	0.9989	0.99942	0.99969	0.99992	0.99997
	PC29	PC30					
Standard deviation	0.02736	0.01153					
Proportion of Variance	0.00002	0.00000					
Cumulative Proportion	1.00000	1.00000					

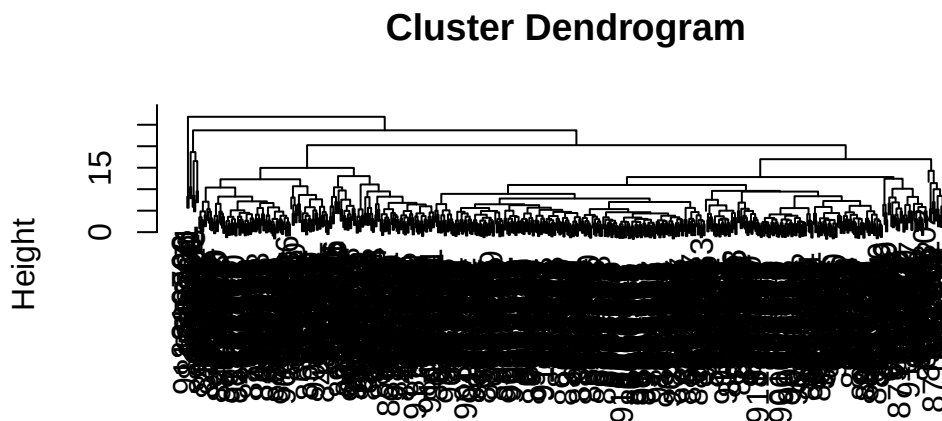
Hierarchical clustering

Just clustering the original data is not very informative or helpful.

```
data.scaled <- scale(wisc.data)
data.dist <- dist(data.scaled)
wisc.hclust <- hclust(data.dist)
```

View the clustering dendrogram result

```
plot(wisc.hclust)
```



```
data.dist
hclust (*, "complete")
```

```
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters)
```

```
wisc.hclust.clusters
  1  2  3  4
177  7 383  2
```

```
table(wisc.hclust.clusters, diagnosis)
```

```

              diagnosis
wisc.hclust.clusters  B  M
1    12 165
2     2   5
3   343  40
4     0   2

```

Results of hierarchical clustering

Clustering on original data was not very productive. The PCA results looked promising. Here we combine these methods by clustering from our PCA results. In other words “clustering in PC space”...

```

##Take the first 3 PCs
dist.pc<- dist(wisc.pr$x[,1:3])
wisc.pr.hclust <- hclust(dist.pc, method="ward.D2")

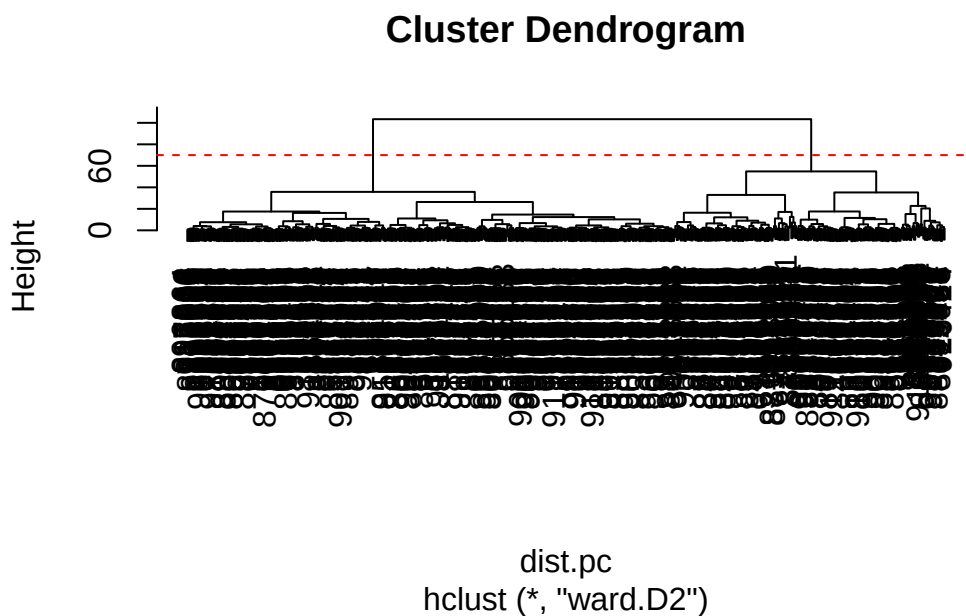
```

Q11. Using the plot() and abline() functions, what is the height at which the clustering model has 4 clusters? View the tree...

```

plot(wisc.pr.hclust)
abline(h=70, col="red", lty=2)

```



To get out clustering membership vector(i.e. our main clustering result) we “cut” the tree at a desired height or to yield a desired number of “k”groups.

```
grps <- cutree(wisc.pr.hclust, k=2)
table(grps)
```

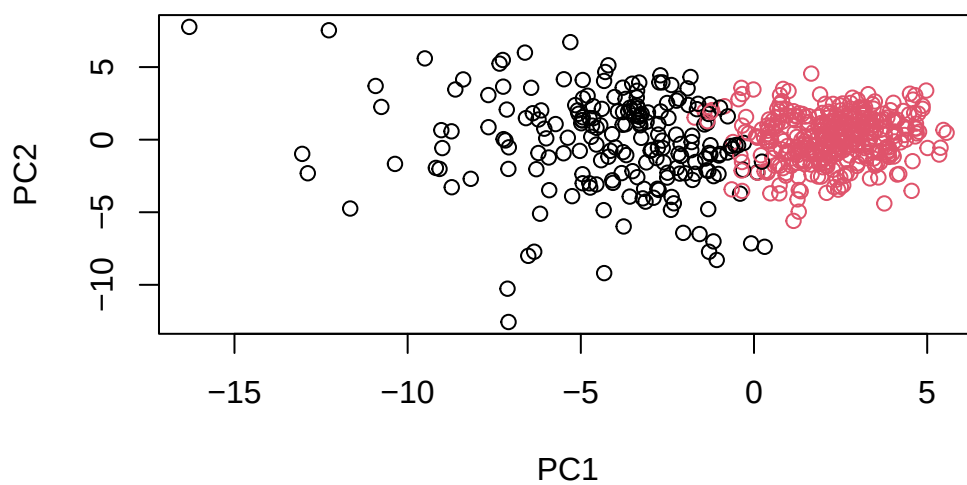
```
grps
  1   2
203 366
```

How does this clustering grps compare to the expert

```
table(grps, diagnosis)
```

```
      diagnosis
grps   B   M
  1  24 179
  2 333  33
```

```
plot(wisc.pr$x[,1:2], col=grps)
```



Q12. Can you find a better cluster vs diagnoses match by cutting into a different number of clusters between 2 and 10?

Cutting the dendrogram at $k = 4$ provides the best trade-off between purity and interpretability (0.905 accuracy)

```
for (k in 2:10) {  
  cat("Number of clusters:", k, "\n")  
  clusters <- cutree(wisc.hclust, k = k)  
  print(table(clusters, diagnosis))  
  cat("\n")  
}
```

Number of clusters: 2

	diagnosis	
clusters	B	M
1	357	210
2	0	2

Number of clusters: 3

	diagnosis	
clusters	B	M
1	355	205
2	2	5
3	0	2

Number of clusters: 4

	diagnosis	
clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

Number of clusters: 5

	diagnosis	
clusters	B	M
1	12	165
2	0	5
3	343	40
4	2	0
5	0	2

Number of clusters: 6

	diagnosis		
clusters	B	M	
1	12	165	
2	0	5	
3	331	39	
4	2	0	
5	12	1	
6	0	2	

Number of clusters: 7

	diagnosis		
clusters	B	M	
1	12	165	
2	0	3	
3	331	39	
4	2	0	
5	12	1	
6	0	2	
7	0	2	

Number of clusters: 8

	diagnosis		
clusters	B	M	
1	12	86	
2	0	79	
3	0	3	
4	331	39	
5	2	0	
6	12	1	
7	0	2	
8	0	2	

Number of clusters: 9

	diagnosis		
clusters	B	M	
1	12	86	
2	0	79	
3	0	3	
4	331	39	
5	2	0	
6	12	0	

7	0	2
8	0	2
9	0	1

Number of clusters: 10

	diagnosis	
clusters	B	M
1	12	86
2	0	59
3	0	3
4	331	39
5	0	20
6	2	0
7	12	0
8	0	2
9	0	2
10	0	1

#Using different methods

Q13. Which method gives your favorite results for the same data.dist dataset?
Explain your reasoning.

```
# Compare linkage methods on the SAME distance matrix
methods <- c("single", "complete", "average", "ward.D2")
hc <- lapply(methods, function(m) hclust(data.dist, method = m))
names(hc) <- methods

# Helper: majority-vote purity for a given hclust and k
purity <- function(h, k) {
  t <- table(cutree(h, k), diagnosis)
  sum(apply(t, 1, max)) / sum(t)
}

# Grid search: purity across k = 2..10 for each method
ks <- 2:10
purity_mat <- sapply(hc, function(h) sapply(ks, function(k) purity(h, k)))
round(purity_mat, 3)
```

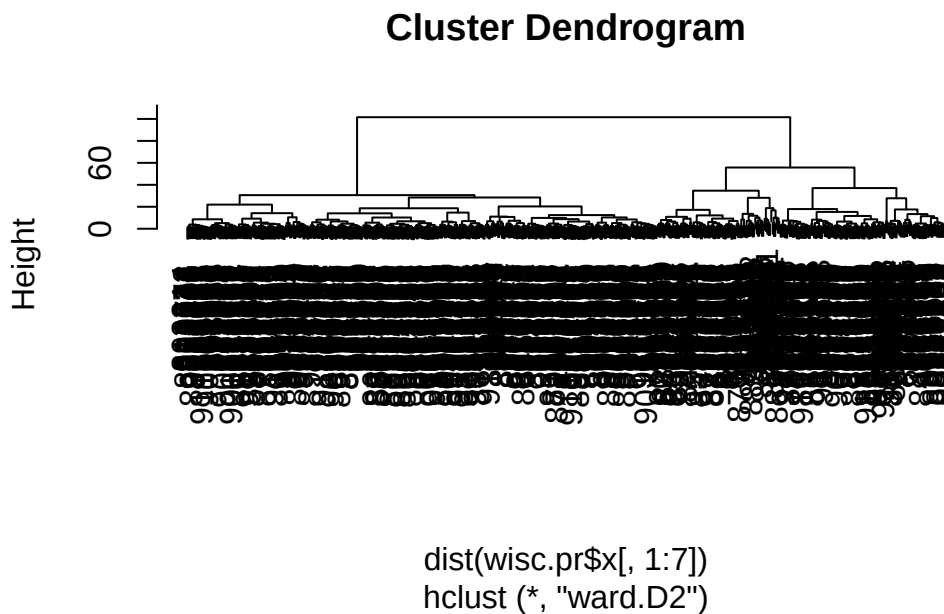
	single	complete	average	ward.D2
[1,]	0.631	0.631	0.633	0.880
[2,]	0.631	0.636	0.633	0.880

[3,]	0.633	0.905	0.633	0.903
[4,]	0.633	0.909	0.634	0.903
[5,]	0.634	0.909	0.645	0.903
[6,]	0.636	0.909	0.715	0.903
[7,]	0.636	0.909	0.715	0.916
[8,]	0.638	0.910	0.715	0.916
[9,]	0.640	0.910	0.715	0.916

I prefer Ward.D2, which achieved the highest purity across most k values (0.88–0.916) and produced clear, compact clusters. While complete linkage also performed well at certain k values, Ward.D2 was more stable overall. By contrast, single linkage had the lowest purity, and average was moderate. This consistency and stability make Ward.D2 the best choice for this dataset.

Combining methods

```
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")
plot(wisc.pr.hclust)
```



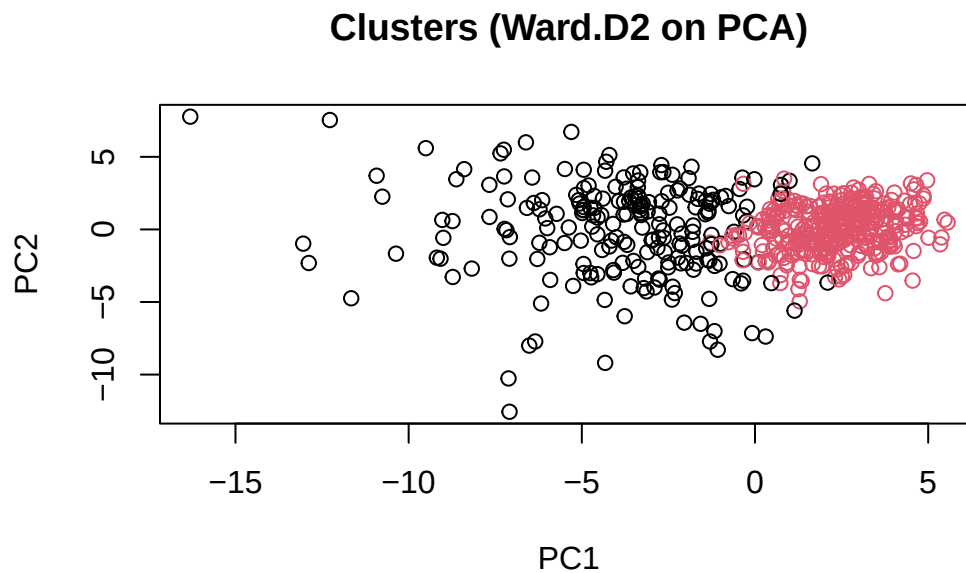
```
# Cut into 2 clusters
grps <- cutree(wisc.pr.hclust, k = 2)
table(grps)
```

```
grps
  1  2
216 353
```

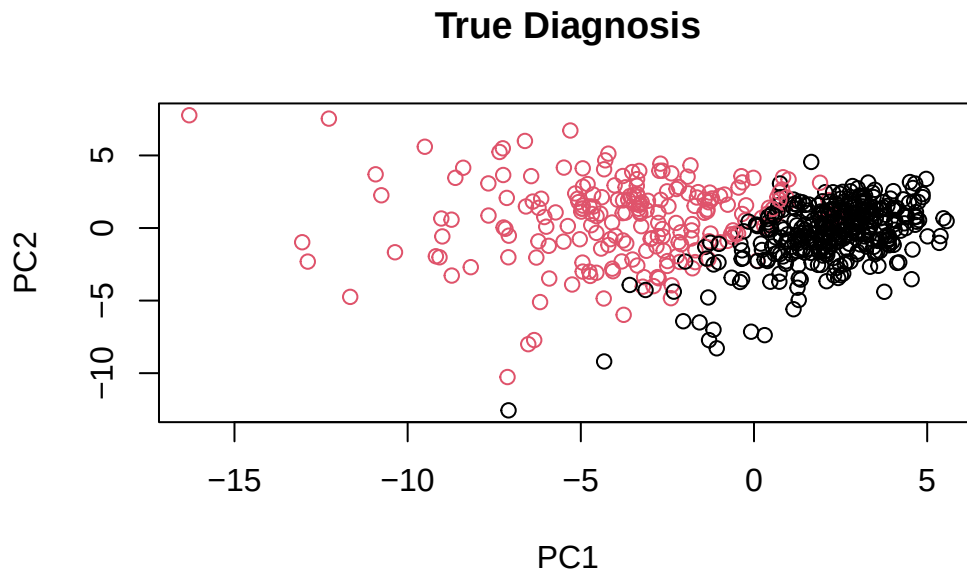
```
table(grps, diagnosis)
```

```
      diagnosis
grps   B    M
  1   28 188
  2  329   24
```

```
plot(wisc.pr$x[,1:2], col = grps, main = "Clusters (Ward.D2 on PCA)")
```



```
plot(wisc.pr$x[,1:2], col = diagnosis, main = "True Diagnosis")
```



Q15. How well does the newly created model with four clusters separate out the two diagnoses?

This is a strong separation between malignant and benign diagnoses with only 2 clusters, showing that PCA improved the cluster structure.

```
# Ward.D2 on the first 7 PCs
wisc.pr.hclust <- hclust(dist(wisc.pr$x[, 1:7]), method = "ward.D2")

# Cut into 2 clusters
wisc.pr.hclust.clusters <- cutree(wisc.pr.hclust, k = 2)

# Compare to diagnosis
table(wisc.pr.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.pr.hclust.clusters	B	M
1	28	188
2	329	24

Q16. How well do the k-means and hierarchical clustering models you created in previous sections (i.e. before PCA) do in terms of separating the diagnoses? Again, use the `table()` function to compare the output of each model (`wisc.km$cluster` and `wisc.hclust.clusters`) with the vector containing the actual diagnoses.

Both k-means and hierarchical clustering separate malignant and benign samples fairly well before PCA. K-means achieved slightly higher purity with just two clusters, while hierarchical clustering needed four clusters for a similar result.

```
# k-means (before PCA)
data.scaled <- scale(wisc.data)
set.seed(1)
wisc.km <- kmeans(data.scaled, centers=2, nstart=50)
table(wisc.km$cluster, diagnosis)
```

	diagnosis	
	B	M
1	343	37
2	14	175

```
# hierarchical (before PCA)
wisc.hclust <- hclust(dist(data.scaled), method="complete")
wisc.hclust.clusters <- cutree(wisc.hclust, k=4)
table(wisc.hclust.clusters, diagnosis)
```

	diagnosis	
wisc.hclust.clusters	B	M
1	12	165
2	2	5
3	343	40
4	0	2

#Sensitivity and Specificity

Sensitivity: $TP / (TP + FN)$ Specificity: $TN / (TN + FP)$

```
# Example: Hierarchical after PCA (k = 2)
tab_pca <- table(wisc.pr.hclust.clusters, diagnosis)

TP <- tab_pca[1, "M"]
FN <- tab_pca[2, "M"]
TN <- tab_pca[2, "B"]
FP <- tab_pca[1, "B"]

sensitivity <- TP / (TP + FN)
specificity <- TN / (TN + FP)
```

```
sensitivity
```

```
[1] 0.8867925
```

```
specificity
```

```
[1] 0.9215686
```

Q17. Which of your analysis procedures resulted in a clustering model with the best specificity? How about sensitivity?

The model with the best specificity was K-means and hierarchical clustering before PCA (both 0.961). The model with the best sensitivity was hierarchical clustering after PCA (0.887). This shows that PCA improved detection of malignant cases, while pre-PCA models were slightly better at identifying benign cases.

Prediction

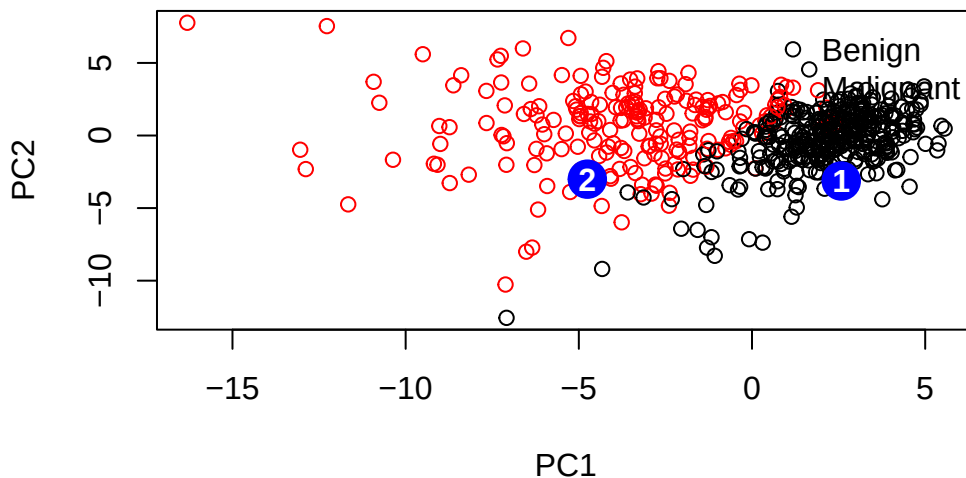
We can use our PCA model for prediction with new input patient sample.

```
url <- "https://tinyurl.com/new-samples-CSV"
new <- read.csv(url)
npc <- predict(wisc.pr, newdata=new)
npc
```

	PC1	PC2	PC3	PC4	PC5	PC6	PC7
[1,]	2.576616	-3.135913	1.3990492	-0.7631950	2.781648	-0.8150185	-0.3959098
[2,]	-4.754928	-3.009033	-0.1660946	-0.6052952	-1.140698	-1.2189945	0.8193031
	PC8	PC9	PC10	PC11	PC12	PC13	PC14
[1,]	-0.2307350	0.1029569	-0.9272861	0.3411457	0.375921	0.1610764	1.187882
[2,]	-0.3307423	0.5281896	-0.4855301	0.7173233	-1.185917	0.5893856	0.303029
	PC15	PC16	PC17	PC18	PC19	PC20	
[1,]	0.3216974	-0.1743616	-0.07875393	-0.11207028	-0.08802955	-0.2495216	
[2,]	0.1299153	0.1448061	-0.40509706	0.06565549	0.25591230	-0.4289500	
	PC21	PC22	PC23	PC24	PC25	PC26	
[1,]	0.1228233	0.09358453	0.08347651	0.1223396	0.02124121	0.078884581	
[2,]	-0.1224776	0.01732146	0.06316631	-0.2338618	-0.20755948	-0.009833238	
	PC27	PC28	PC29	PC30			
[1,]	0.220199544	-0.02946023	-0.015620933	0.005269029			
[2,]	-0.001134152	0.09638361	0.002795349	-0.019015820			

```
cols <- ifelse(diagnosis == "B", "black", "red")

plot(wisc.pr$x[,1], wisc.pr$x[,2], col = cols, pch = 1,
     xlab = "PC1", ylab = "PC2")
points(npc[,1], npc[,2], pch = 21, bg = "blue", col = "blue", cex = 2.6)
text(npc[,1], npc[,2], labels = 1:2, col = "white", font = 2)
legend("topright", c("Benign", "Malignant"), col=c("black", "red"), pch=1, bty="n")
```



Q18. Which of these new patients should we prioritize for follow up based on your results?

Patient 1 should be prioritized for follow-up because their projected PCA coordinates place them in or near the malignant cluster, which is associated with higher cancer risk. Patient 2 falls in the benign region and is lower priority, though should still be monitored as appropriate.